

CareLink Follow-Up

A reliable missed-appointment worklist for connected and offline clinics

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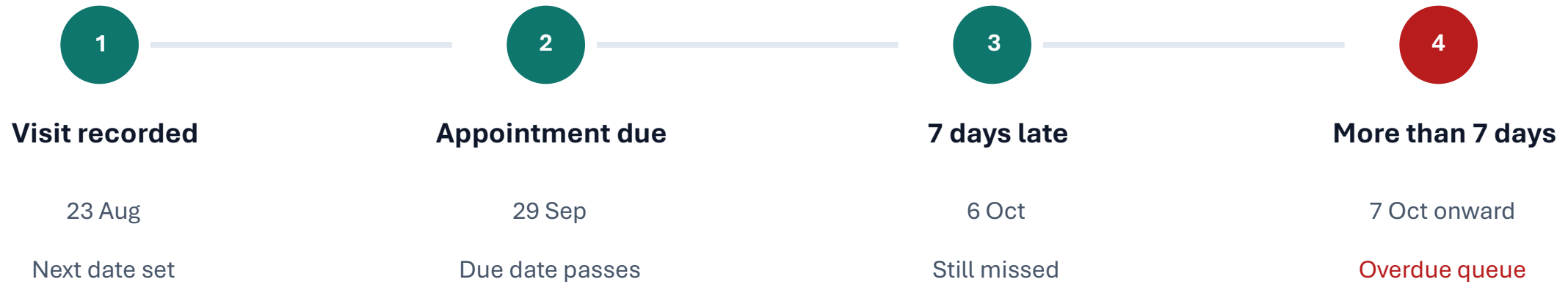
Software Engineer technical assessment

7th October 2026

Follow-up rule and assumptions

02

Latest visit controls the current status

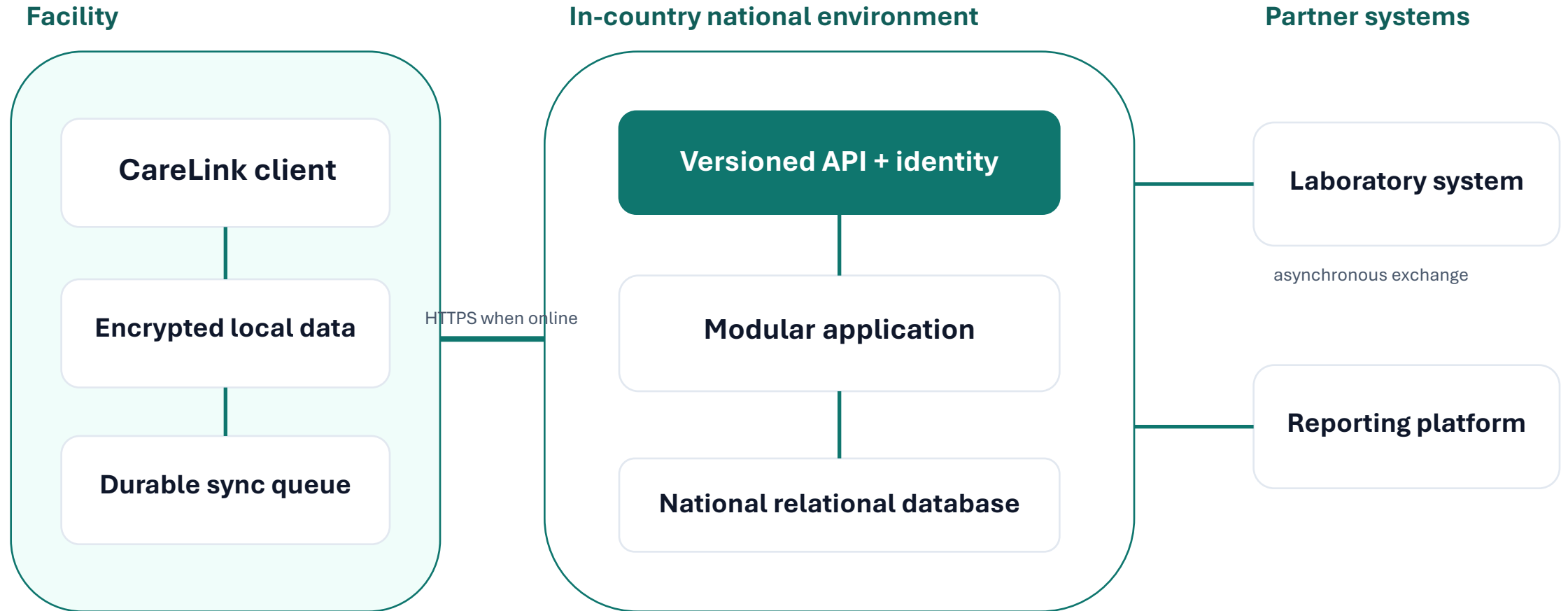


A later attended visit supersedes an earlier missed appointment.

The worklist represents the patient's latest actionable status, not every historical missed date.

Proposed national architecture

03



Working vertical slice

04

 **CareLink**
Continuity of care

Demonstration environment

Access profile CareLink manager · all facilities

Patient follow-up

Review missed appointments and coordinate patient outreach.

 **2**
Patients in this queue
After filters are applied

 **FAC-0101**
Current facility
Mwansa Urban Clinic

 **7 days**
Overdue threshold
Strictly beyond this window

Queue controls
Filter patients

Facility ID

FAC-0101

Follow-up status

Overdue

Threshold in days

7

Rows per page

50

Sort order

Most overdue first

Action queue
Overdue follow-ups

1-2 of 2

Patient	Appointment	Priority	Contact
MB Mary Banda 0101-000001	22 Sept 2026 15 days late	Overdue	+260 97 000 0000 No contact attempt
JP John Phiri 0101-000002	29 Sept 2026 8 days late	Overdue	+260 97 000 0000 No contact attempt

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React + TypeScript

Typed service, explicit states and keyboard-operable filters

.NET 8 API

Validation, access scope, correlation IDs and database pagination

SQLite

Migration plus readable and assessment-scale seeders

LIVE DEMO

Swagger authentication to boundary filters to React error/retry path

Engineering decisions and trade-offs

05

Modular monolith

chosen

Microservices

alternative

Lower operational cost for a nine-engineer team; modules preserve a later extraction path.

Local facility data

chosen

Online-only client

alternative

Clinic work continues during outages; sync and conflict handling become explicit responsibilities.

Database-side query

chosen

Load then filter in C#

alternative

Indexes, filters and pagination limit memory, bandwidth and records rendered in the browser.

Server access scope

chosen

Hide facilities in React

alternative

API enforcement prevents a user bypassing the interface and requesting another facility directly.

Code review: changes required before merge

06

1

SQL injection and embedded database credentials

Risk: cross-facility disclosure, data alteration and an unrotatable privileged secret.

2

Facility and role supplied by the caller

Risk: a caller can request another facility unless the API uses verified claims and scope.

3

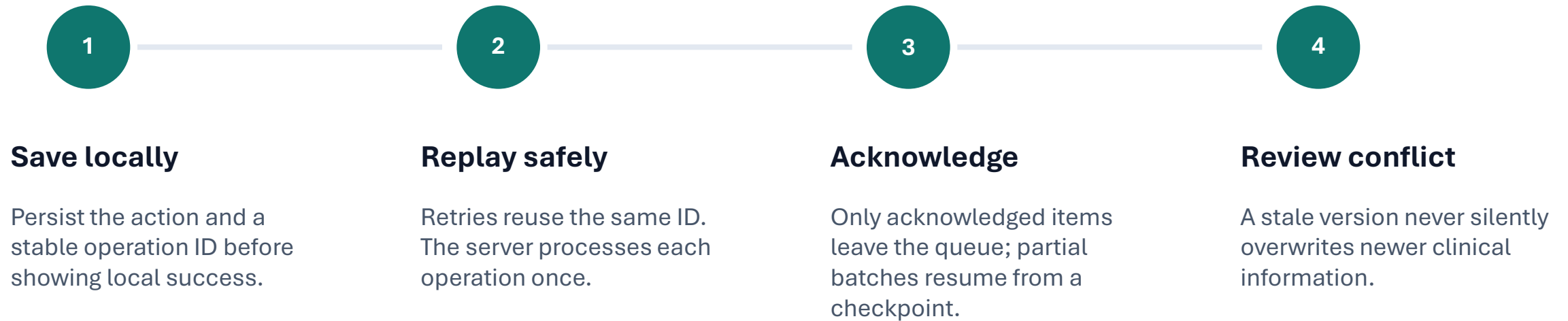
Partial writes and false success

Risk: staff believe an encounter or contact saved when the server rolled back or never received it.

Focused refactor: parameters + trusted identity + one transaction + visible failure

Offline replay and conflict handling

07



Existing duplicates remain clinical records

A data clerk reviews candidates. Confirmed identities are linked or merged with original identifiers and an audit trail preserved.

Evidence, limitations and next iterations

100,000

patients seeded

400,000

visits seeded

35 ms

median warm query

16 / 16

automated tests passed

Current limitations

- Static assessment tokens, not production identity
- Offline sync and partner integrations are designs only
- Contacted POST left as a documented stretch goal

With future version

- Implement authorised, idempotent contact recording
- Add production-engine integration and concurrent load tests